**SIMATS ENGINEERING**

**ASSESSMENT TOOL 2**

**COURSE CODE: MLA03**

**COURSE NAME: Reinforcement learning**

**COURSE OUTCOME COVERED**

***CO4: Evaluate model-based reinforcement learning approaches for prediction, planning, and policy optimization. (BL3, BL4, BL5)***

*Assessment Tool Weightage: 40%*

**Dr G. A. SENTHIL**

**QUESTIONS: 25   Total Marks: 50**

**Annexure A**

**Design and Develop Model**

**DURATION: 4 HRS (2 Sessions)**

*Note: The following five questions have been selected and answered from the original question bank of twenty-five questions, in accordance with the requirements of this assessment.*

***Code of Conduct:***

*I ___A Shiva Shankar Reddy(192425174_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

*I understand that any violation of academic integrity rules will result in disciplinary action.*

***Signature of the student:***

*A Shiva Shankar Reddy*

## Model-Based RL Approach

**Question 1: Draw a flowchart illustrating the complete workflow of a Model-Based Reinforcement Learning agent from environment interaction to policy optimization.** ***(5 Marks)***

**Answer:**

A Model-Based Reinforcement Learning (MBRL) agent operates in a closed loop that begins with interaction with the real environment and ends with the execution of an optimized action. The agent first interacts with the environment and records transition tuples of the form (state, action, reward, next state). This experience is used to learn or continually update an internal environment model that approximates the true transition and reward functions. Once a sufficiently accurate model is available, a planning module uses it to simulate many possible future trajectories without requiring additional real-world interaction. The outcomes of these simulated rollouts are then used to optimize the policy, typically by maximizing expected cumulative reward. The optimized policy selects an action, which is executed in the real environment, and the resulting new experience is fed back into the loop, allowing the model and policy to be refined continuously. The complete workflow is illustrated below.

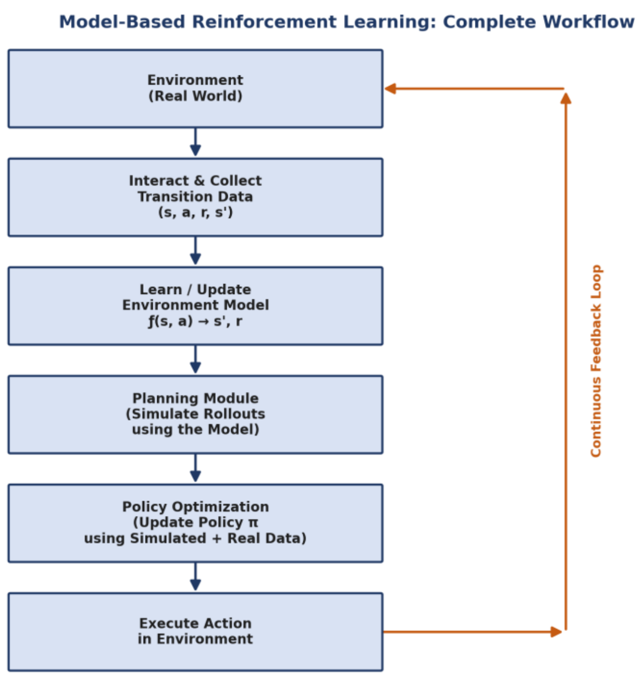


*Figure 1: Complete workflow of a Model-Based Reinforcement Learning agent.*

This closed-loop design is what gives model-based methods their key advantage: because planning is performed inside a learned simulator, the agent needs far fewer real environment interactions to reach a good policy than purely model-free approaches.

**Question 5: Prepare a concept map comparing Model-Based Reinforcement Learning and Model-Free Reinforcement Learning.** ***(5 Marks)***

**Answer:**

Both Model-Based and Model-Free Reinforcement Learning are branches of the broader Reinforcement Learning paradigm, but they differ fundamentally in how they use experience to improve behaviour. Model-Based RL explicitly learns an approximate model of the environment's dynamics and uses it for internal planning and simulated rollouts, which generally makes it far more sample-efficient. Model-Free RL, by contrast, learns a policy or value function directly from real trajectories without ever constructing an explicit model, which tends to be simpler to implement and more robust to model errors, but typically requires substantially more environment interaction to converge. The concept map below summarizes these relationships.

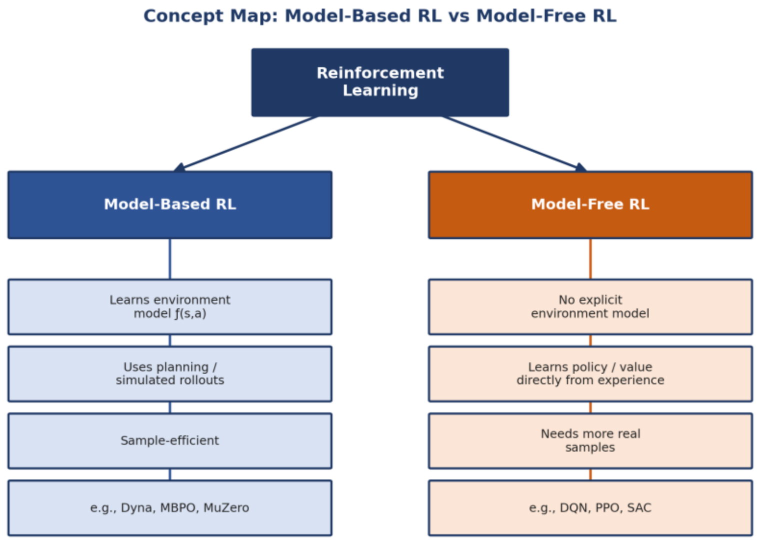


*Figure 2: Concept map comparing Model-Based RL and Model-Free RL.*

In practice, hybrid approaches such as MBPO combine the sample efficiency of model-based planning with the asymptotic performance and simplicity of model-free policy updates, offering a practical middle ground between the two paradigms.

## Sampling-Based Planning

**Question 11: Draw a flowchart illustrating the sampling-based planning process in Model-Based Reinforcement Learning.** *(5 Marks)*

**Answer:**

Sampling-based planning is a widely used technique in Model-Based RL, particularly in Model Predictive Control (MPC) settings. Starting from the current state, the planner samples a large number of candidate action sequences. Each candidate sequence is simulated forward using the learned environment model to generate a predicted trajectory of future states and rewards. These simulated trajectories are then evaluated, typically by summing the predicted (and often discounted) rewards along the trajectory. The action sequence that yields the highest predicted return is selected, and only its first action is executed in the real environment, following the receding-horizon principle. After execution, the agent observes the new state and repeats the entire sampling and evaluation process, allowing it to continuously re-plan in response to new information. This process is depicted in the flowchart below.

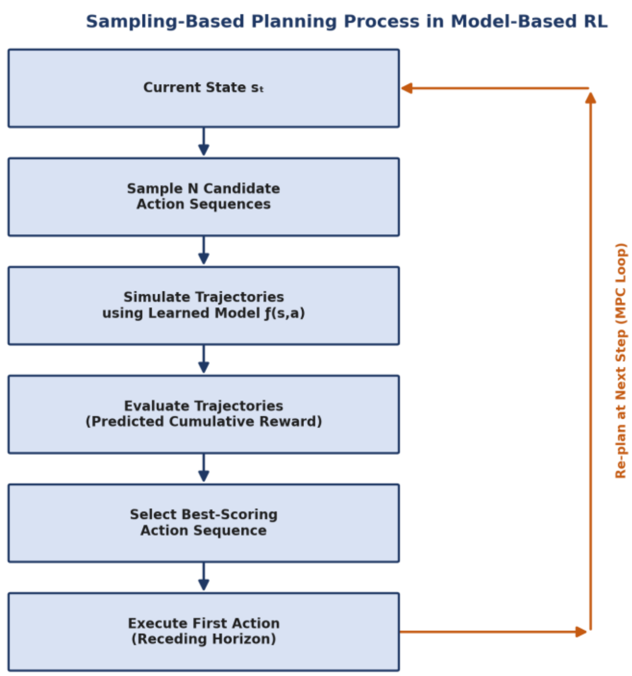


*Figure 3: Sampling-based planning process in Model-Based Reinforcement Learning.*

Because only the first action of the best sequence is executed before re-planning, sampling-based planning is naturally robust to small model inaccuracies and to unexpected changes in the environment.

## Model-Based Data Generation

**Question 18: Draw a block diagram explaining model-generated experience replay in Reinforcement Learning.** *(5 Marks)*

**Answer:**

Model-generated experience replay augments the standard replay buffer used in Reinforcement Learning with synthetic transitions produced by a learned environment model, in addition to the real transitions collected by interacting with the environment. The real environment supplies genuine experience, which is used both to populate the replay buffer directly and to train the environment model. Once trained, the environment model is used to generate additional simulated rollouts, which are also added to the replay buffer. The policy or value network is then updated by sampling mini-batches from this combined pool of real and synthetic experience, which greatly increases the effective amount of training data available without requiring additional costly interaction with the real environment. The resulting improved policy is subsequently used to collect better-quality real experience, closing the loop. This mechanism is illustrated in the block diagram below.

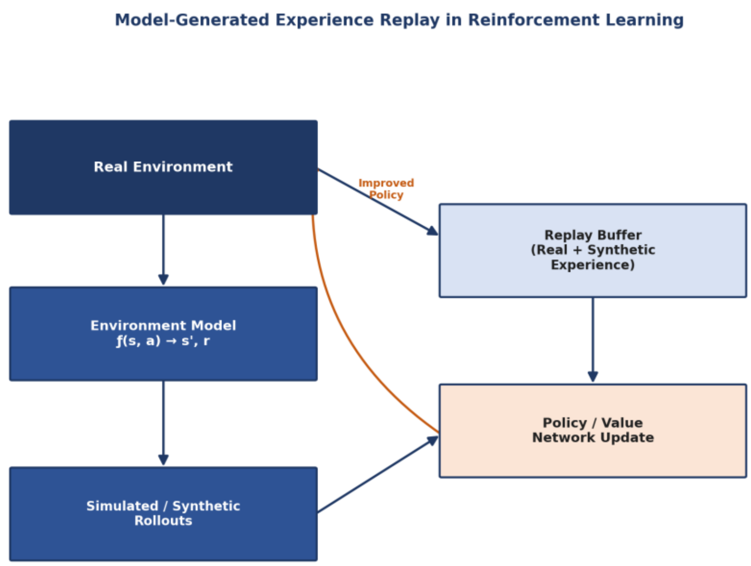


*Figure 4: Model-generated experience replay mechanism.*

This technique substantially improves sample efficiency, since every unit of real-world interaction can be leveraged many times over through model-generated rollouts, at the cost of some risk of compounding model error if the learned dynamics model is inaccurate.

## Value-Equivalence Prediction and Policy Optimization

**Question 23: Draw a block diagram illustrating the complete Model-Based Policy Optimization (MBPO) framework.** *(5 Marks)*

**Answer:**

Model-Based Policy Optimization (MBPO) is an algorithm that interleaves real-world data collection, dynamics-model learning, short model-based rollouts, and model-free policy updates. The agent begins by collecting real experience from the environment and uses it to train a probabilistic dynamics model that captures both the predicted next state and the associated uncertainty. This model is then used to generate many short rollouts branching from previously visited real states, which limits the compounding of model error over long horizons. The resulting model-generated transitions are combined with the real transitions in a shared data pool, and a model-free algorithm such as Soft Actor-Critic (SAC) is used to update the policy using this combined data. The improved policy is then deployed back into the real environment to collect fresh data, and the entire cycle repeats iteratively. The complete framework is shown below.

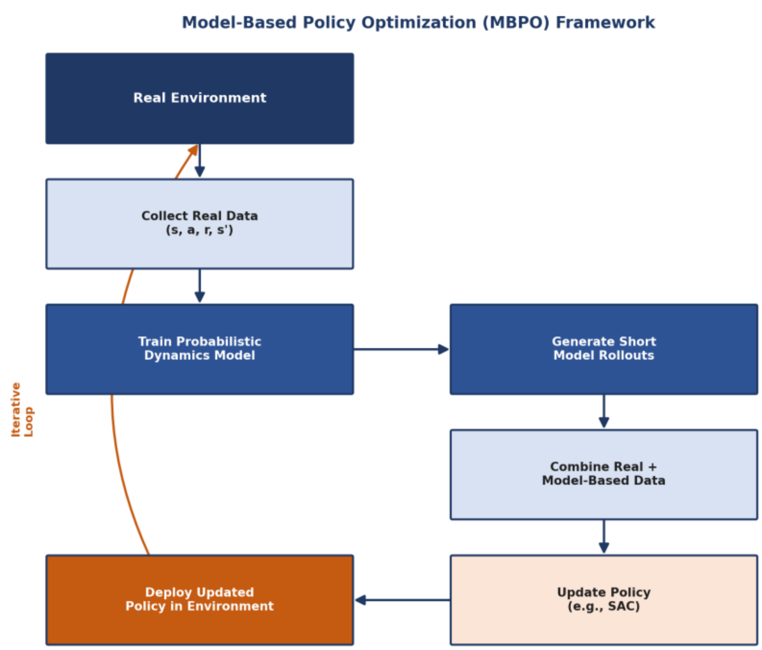


*Figure 5: Model-Based Policy Optimization (MBPO) framework.*

By restricting rollouts to short horizons starting from real states, MBPO achieves sample efficiency close to that of pure model-based methods while retaining the strong asymptotic performance typically associated with model-free algorithms.

## RUBRICS FOR CALCULATION

| Criteria | Weightage | Level 4 – Exemplary | Level 3 – Proficient | Level 2 – Developing | Level 1 – Beginning |
|---|---|---|---|---|---|
| Technical Content Accuracy | 30 | Highly accurate, in-depth, and insightful technical explanation | Technically correct content with adequate depth and clarity | Basic concepts presented with limited accuracy and depth | Content contains major technical errors; concepts poorly understood |
| Problem Identification | 25 | Problem rigorously analyzed using appropriate and advanced methods | Problem clearly defined with a logical engineering approach | Problem identified but analysis or methodology is weak | Problem statement unclear or incorrect; no logical approach |
| Use of Tools | 20 | Advanced tools, well-analyzed data, and highly effective visuals | Appropriate tools and data used effectively with clear visuals | Limited use of tools and basic data interpretation | Minimal or incorrect use of tools and data; poor visuals |
| Communication | 15 | Highly engaging, confident, and audience-focused delivery | Clear, organized, and professional communication | Understandable but lacks clarity, structure, or confidence | Poor organization, unclear speech, ineffective slides |
| Professionalism | 10 | Exemplary professionalism, leadership, and ethical awareness | Professional conduct with effective team collaboration | Basic professionalism; uneven team participation | Lacks professionalism; minimal contribution or ethical awareness |